Instructions for Sharing HydroSOS Data via SharePoint

UK Centre for Ecology & Hydrology (UKCEH), March 2026

This document outlines how to share HydroSOS products with UKCEH for inclusion in the Global HydroSOS Portal. This is intended for data providers who do not have an API that UKCEH can access and whilst testing for WHOS data transfer for HydroSOS products takes place.

*Please contact **wmohydrosos@ceh.ac.uk** if you have any questions about these instructions.*

Accessing the SharePoint site

To access the SharePoint site, you will need a Microsoft account with two-factor authentication (2FA) enabled via an authentication app.

Note: If you do not have an existing Microsoft account, you should be able to use your existing (i.e. Microsoft) email address to register for a Microsoft account (see below).

STEP 1. Accessing the SharePoint site with an existing Microsoft account

If you have an existing Microsoft account, you can use this to log in to the SharePoint site.

OR

STEP 1. Accessing the SharePoint site with NEW Microsoft account

- If you **DO NOT** have an existing Microsoft account, register for a free account here: https://account.microsoft.com/account/Account
- On the logon screen find the button at the bottom marked **No account, Create One!**
- On the Create account page, select **Use your email address instead.**
- Type in the email address you wish to use and set a password for the Microsoft account (this password will remain separate to the password used for the email).
  - Follow the rest of the steps to finish creating the account. You will receive an email from Microsoft asking you to verify the account. Once done, this will allow you use the pre-existing email as a Microsoft account.

STEP 2. Steps to use two-factor authentication with your Microsoft account

Follow the link to use two-factor authentication with your Microsoft account:

https://support.microsoft.com/en-gb/account-billing/how-to-use-two-stepverification-with-yourmicrosoft-account-c7910146-672f-01e9-50a0-93b4585e7eb4

STEP 3. Request membership

1. Before your first data upload, **check you have access** to the SharePoint site.

2. If you do not have access, please contact the SharePoint administrator to be added as a member.

- You can do this via the site directly by **submitting the request access box**, noting your name, organisation and country, **OR** by emailing wmohydrosos@ceh.ac.uk.
- We will approve and get back to you as soon as possible.

Expected data files

1. Status and Outlook Data

The SharePoint site can only be populated with .csv, .json and .geotiff files of monthly status and outlook data.

The files should either be populated with the HydroSOS category (i.e. 1 to 5, where 1 is Notably low and 5 is Notably high) or as percentiles of 0 to 100.

2. Metadata

One metadata .csv file should be included for each variable that is uploaded. The metadata file should be named **"station_metadata.csv"**. An example metadata file can be found here and should include at least the following three columns (with these column names in [square brackets]):

1. Station ID [station_id]

2. Latitude [lat] (in WGS84 coordinate system in decimal degrees)

3. Longitude [lng] (WGS84 coordinate system in decimal degrees)

It would also be appreciated you can include these columns:

4. Catchment area if known, in km^2 [catchment_area] (if data are catchment based, remove if not relevant, e.g. for groundwater levels or soil moisture data)

5. Record start date (for the station, format: YYYY-MM-DD, note this may differ from the standard period used) [record_start_date]

6. River name [river_name]

7. Standard reference period start date [ref_start] (YYYY-MM-DD) (default 1991-01-01)

8. Standard reference period end date [ref_end] (YYYY-MM-DD) (default 2020-01-01)

9. Percentage missing (percentage of missing daily data in the reference period used to calculate the HydroSOS product; value between 1 and 100) [missing_data_pcent]

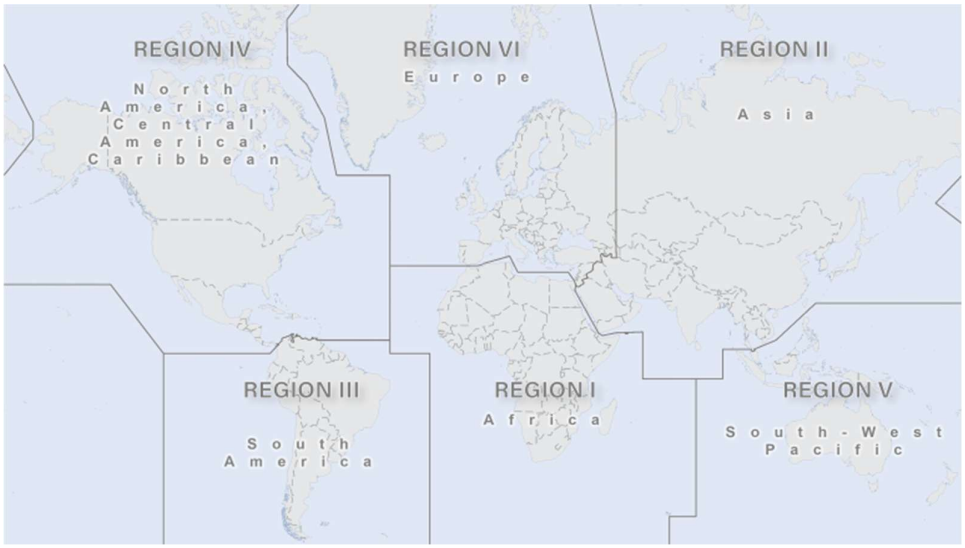
10. Data provider name [provider_name]

11. Data publisher name [publisher_name]

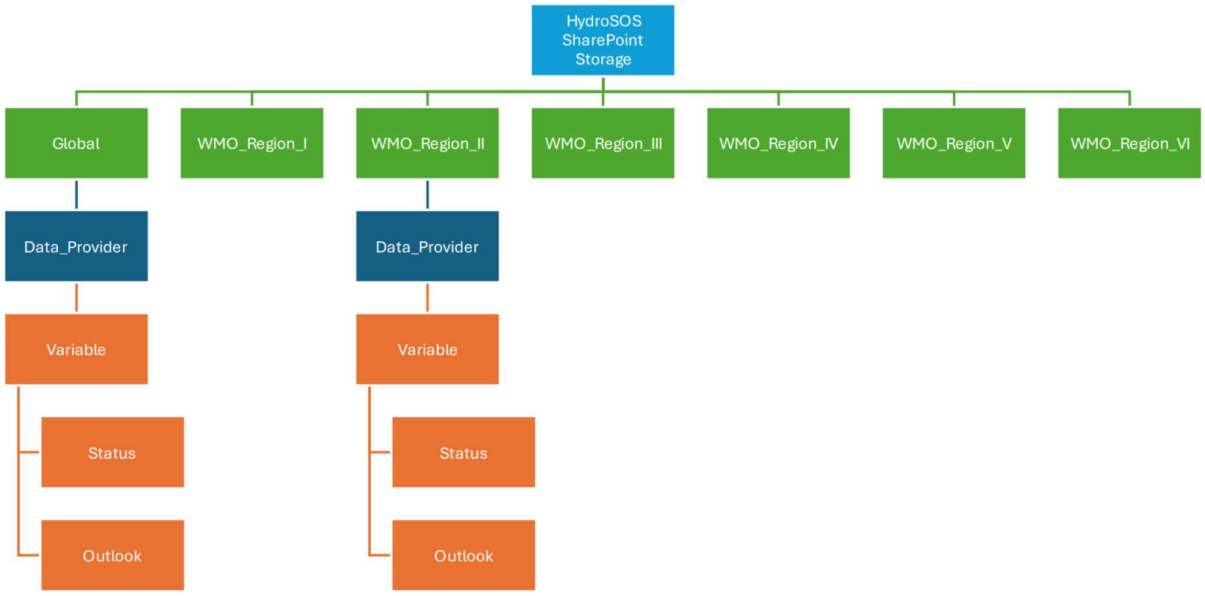
A	B	C	D	E	F	G	H	I	J	K
station_id	lat	lng	catchment_area	record_start_date	river_name	ref_start	ref_end	missing_data_pcent	provider_name	publisher_name
A	57.2321	-1.2324								
B	54.3214	56.3213								
C	-32.3213	-21.3245								

Uploading Data

Data in the HydroSOS SharePoint site are organised in the first instance by either Global or by WMO Regional Association data providers. An example folder structure is shown in the flow chart below. There is also an example directory under Documents → {WMO_Region} on the SharePoint site.



Find out which **WMO Region** your country belongs to here.



1. Navigate to the **Documents** directory.

2. Locate the appropriate directory: **Documents→ WMO_Region_I→ Variable→ etc...**

   - If you are a global data provider, create the relevant sub-folders in the Global data directory

3. If necessary, create sub-folders following the expected structure under the relevant WMO Region directory:

   - WMO_Region_X → {Data_Provider} → {Variable Name} → {Status) → {counts}
   - (WMO_Region_X → {Data_Provider} → {Variable Name} → {Outlook) → {YYYY-MM} → {accumulated → counts} / {single → counts}

• Replace names in curly brackets with the relevant information.

• Data_Provider should be a **lowercase acronym** for your organisation (with no spaces).

• Variable name should be one of: **flow, runoff, groundwater_level, soil_moisture, swe, rainfall or water_level**

• Do not use spaces in the names of any folders that you create (or files you upload), use underscore instead (i.e., _) and use all lowercase letters.

4. Upload the **CSV timeseries files** into the corresponding folder, i.e. status data goes into the status/counts/ folder and the outlook data goes into the outlook folder- outlooks/YYYY-MM/ single/counts or outlooks/YYYY-MM/accumulated/counts

Status data files:

- Status files should be named as cat_{stationID}.csv e.g., cat_1001.csv
- An example status file can be found here
- The file should include the date as YYYY-MM, the category as 1 to 5 and optionally, a flag which notes whether the data used to calculate the status category is provisional or not (where 1 = data have been quality controlled and won't change.

0 = unchecked and is liable to change):

date	category	QA
2024-04	3	1
2025-05	5	0

5. In the same folder for **status or outlook**, please also upload **.json files** which hold the data about all the stations for a particular month. i.e. the same data as the timeseries files but transposed for a given month. This is used to colour up the station markers on the map.

The files should be named as the month, i.e., {YYYY-MM}.json (e.g. 2020-05.json).

Example status json data:

```
[{"category":3,"stationID":"11042"},{"category":5,"stationID":"
1-Deim"},{"category":5,"stationID":"270001"}]
```

Example outlook json data:

```
[{"stationID": "55021", "low": 17, "norm": 30, "high": 15},
{"stationID": "29009", "low": 16, "norm": 30, "high": 16}]
```

For global data, instead of .json files, **Geotiffs of gridded data aggregated to Hydrobasins Level 4** regions are the expected output, which will be served up on the portal as a web map service (WMS). See the Global → OUTLAST folder for an example of the required data structure.

Outlook data files:

- Outlook files should be named as {stationID}_counts.csv.
- An example outlook csv data is below.
- The **single/accumulated** file should include the date and the count of ensemble members that are in each of the categories (notLow is notably low, belNorm is below normal, norm is normal, abNorm is above normal, notHigh is notably high):

date	notLow	belNorm	Norm	abNorm	notHigh
2024-02	0	0	0	0	2
2024-03	0	0	0	0	2
2024-04	0	0	0	0	2
2024-05	0	0	0	1	1
2024-06	0	0	2	0	0
2024-07	0	1	1	0	0
2024-08	1	1	0	0	0

Final Pre-Upload Checklist (Mandatory)

Before uploading any new data or stations to the HydroSOS SharePoint site, please carefully check the following. Failure to follow these steps will prevent the Portal from updating correctly.

A. File Format – Exact Match Required

☐ All files are in the correct format (.csv, .json, .geotiff only).
☐ Column names match exactly those specified in this document (including lowercase, underscores, and spelling).
☐ Dates follow the correct format:

- Status files: YYYY-MM
- Metadata: YYYY-MM-DD

☐ Categories are coded correctly:

- Status: integers 1–5 only
- Percentiles (if used): values 0–100 only

☐ Folder names contain:

- No spaces
- Lowercase letters only
- Underscores (_) instead of spaces
  ☐ File names follow the exact naming convention (e.g. cat_1001.csv, 2020-05.json, {stationID}_counts.csv).

Important: Even small differences (e.g. capital letters, extra spaces, different column names) will prevent the system from reading the files correctly.

B. Metadata Requirements

☐ A station_metadata.csv file is included.
☐ All required columns are present:

- station_id
- lat
- lng

☐ Station IDs in metadata match exactly the station IDs used in the data files.
☐ Latitude/longitude are provided in WGS84 decimal degrees.
☐ Reference period dates are correctly specified (if not using default 1991–2020).
☐ Any newly added stations are included in the metadata file.

Missing or inconsistent metadata will prevent stations from appearing on the Portal.

C. Adding New Stations – Manual Step Required

If you add new stations to SharePoint, you MUST notify UKCEH/WMO.

The Portal requires a manual configuration step before new stations can appear.

☐ You have emailed wmohydrosos@ceh.ac.uk to notify us of:

- New station IDs
- Variable type
- Region and data provider folder location

Without this notification and manual configuration, newly uploaded stations will NOT appear on the HydroSOS Portal, even if files are correctly uploaded.

D. Data Quality Check

☐ Raw data have been quality controlled before generating HydroSOS products.
☐ Dates have been checked for errors or misalignment.
☐ Files have been placed in the correct organisation and variable directory.
☐ You have reviewed one file after upload to ensure formatting has not changed.

E. After Upload

☐ You have informed UKCEH/WMO that the upload is complete.
☐ You have indicated whether:

- This is a routine update, OR
- New stations have been added.

Important Reminder

The HydroSOS SharePoint system does not automatically detect structural changes. Uploads that do not strictly follow these instructions will not be processed correctly.

If in doubt, please contact **wmohydrosos@ceh.ac.uk** before uploading.

Additional Notes:

- Please note, unfortunately UKCEH IT Policy does not enable us to provide programmatic write access to the SharePoint site. Files should be manually copied to the relevant directories.
- Please verify data integrity before uploading.
- UKCEH will not verify any data uploaded to the SharePoint site. Although if there are any issues with the data, we will get back to you.
- Code to produce the .csv, .json and .geotiff files is available: GitHub - https://github.com/wmo-im/HydroSOS

If you encounter issues, reach out to our support email address: **wmohydrosos@ceh.ac.uk**

By following these guidelines, we can maintain an organised and efficient data-sharing system for HydroSOS. Thank you for your cooperation!